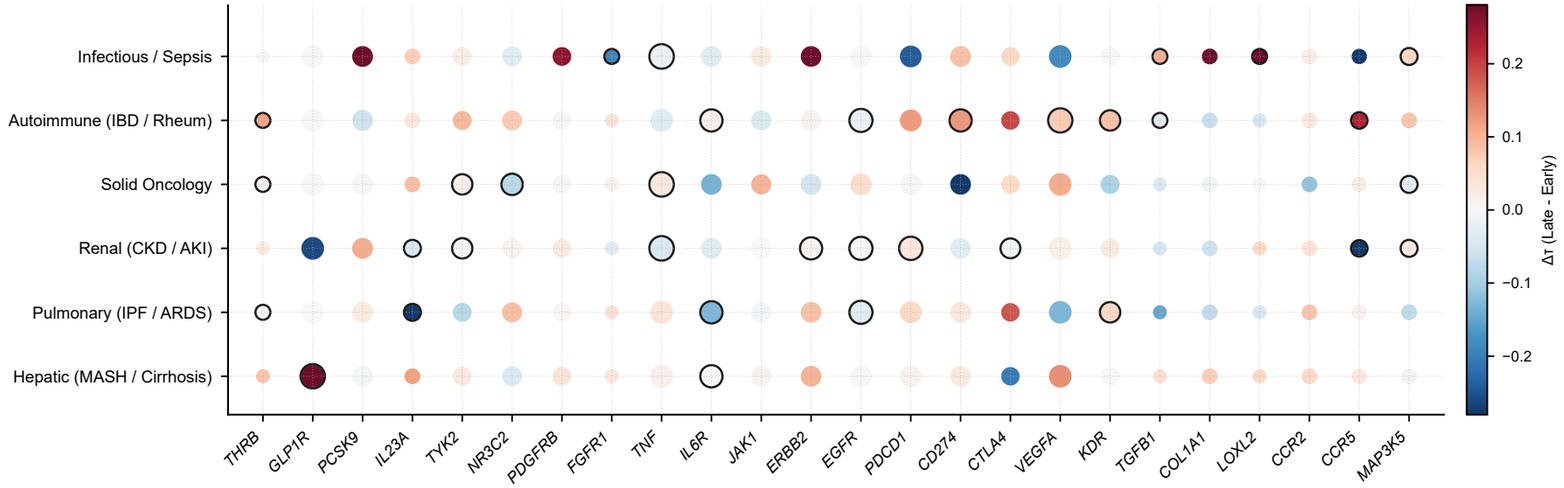
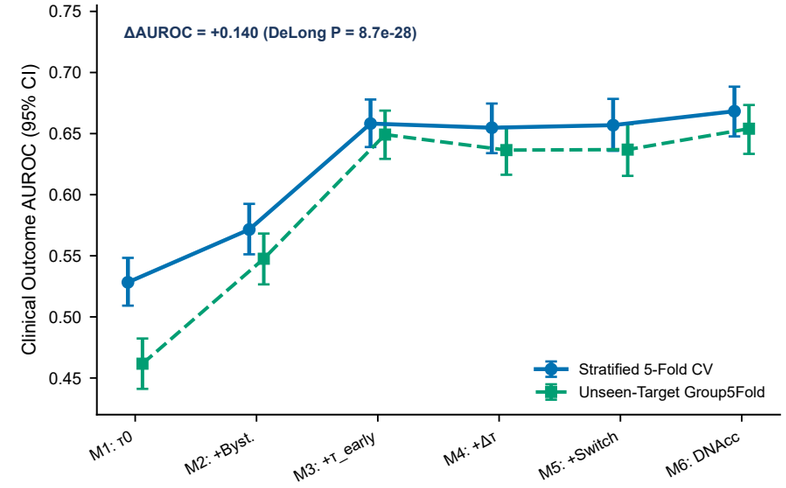
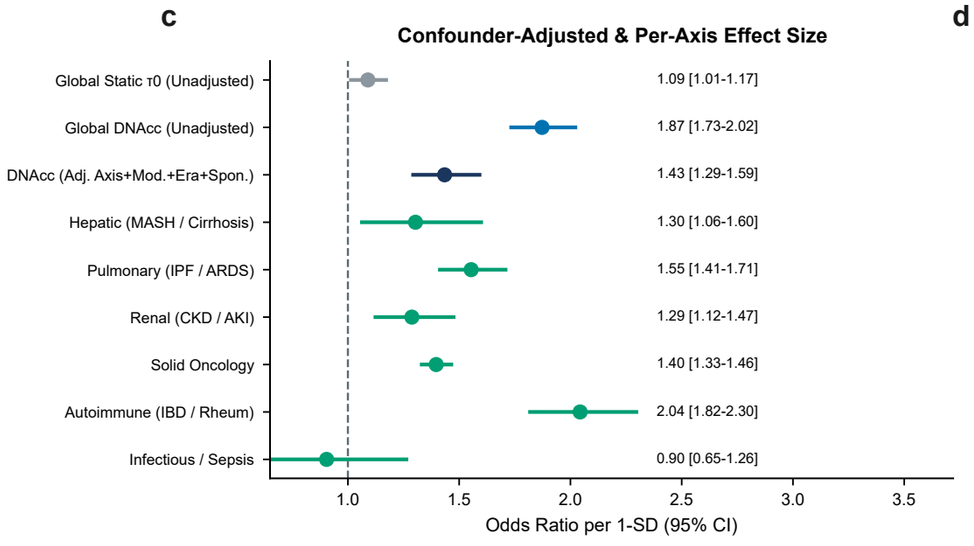
a


Single-Cell Target Specificity Drift (Δτ) & Ectopic Lineage Switch (Black Ring)
Infectious / Sepsis
Autoimmune (IBD / Rheum)
Solid Oncology
Renal (CKD / AKI)
Pulmonary (IPF / ARDS)
Hepatic (MASH / Cirrhosis)
THRB
GLP1R
PCSK9
IL23A
TYK2
NR3C2
PDGFRB
FGFR1
TNF
IL6R
JAK1
ERBB2
EGFR
PDCD1
CD274
CTLA4
VEGFA
KDR
TGFB1
COL1A1
LOXL2
CCR2
CCR5
MAP3K5
0.2
0.1
0.0
−0.1
−0.2
Δτ (Late - Early)

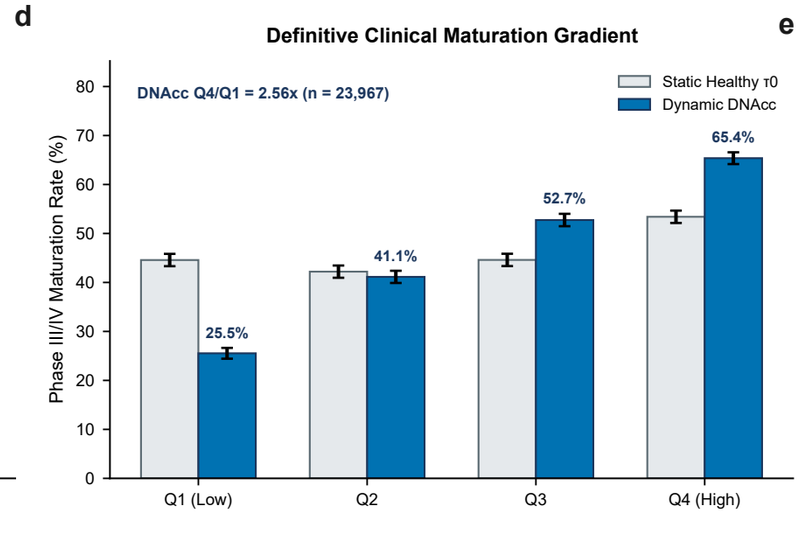
b


6-Step Orthogonal DNAcc Ablation (n = 11,843)
ΔAUROC = +0.140 (DeLong P = 8.7e-28)
Clinical Outcome AUROC (95% CI)
0.75
0.70
0.65
0.60
0.55
0.50
0.45
Stratified 5-Fold CV
Unseen-Target Group5Fold
M1: τ0
M2: +Byst.
M3: +τ_early
M4: +Δτ
M5: +Switch
M6: DNAcc

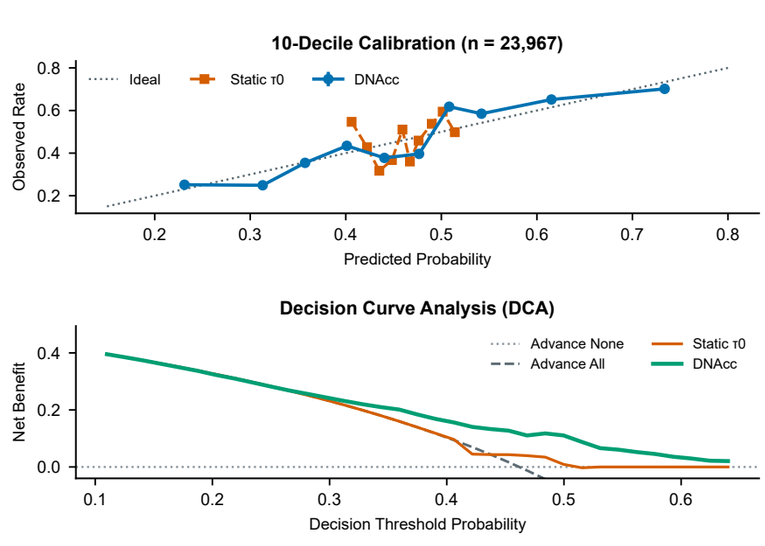
c


Confounder-Adjusted & Per-Axis Effect Size
Global Static τ0 (Unadjusted)
Global DNAcc (Unadjusted)
DNAcc (Adj. Axis+Mod.+Era+Spon.)
Hepatic (MASH / Cirrhosis)
Pulmonary (IPF / ARDS)
Renal (CKD / AKI)
Solid Oncology
Autoimmune (IBD / Rheum)
Infectious / Sepsis
1.09 [1.01-1.17]
1.87 [1.73-2.02]
1.43 [1.29-1.59]
1.30 [1.06-1.60]
1.55 [1.41-1.71]
1.29 [1.12-1.47]
1.40 [1.33-1.46]
2.04 [1.82-2.30]
0.90 [0.65-1.26]
1.0
1.5
2.0
2.5
3.0
3.5
Odds Ratio per 1-SD (95% CI)

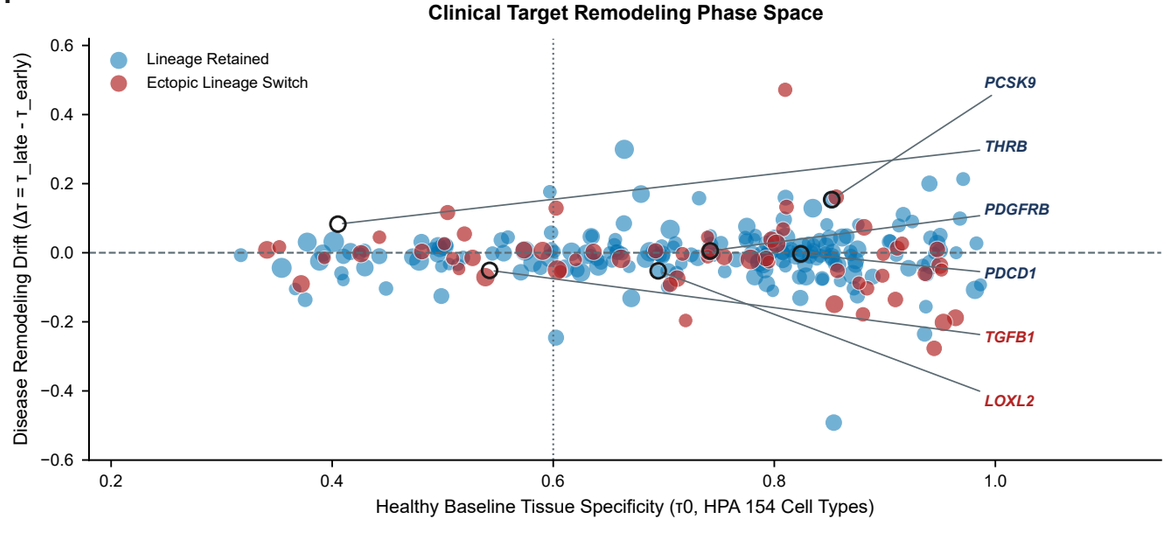
d


Definitive Clinical Maturation Gradient
DNAcc Q4/Q1 = 2.56x (n = 23,967)
Static Healthy τ0
Dynamic DNAcc
Phase III/IV Maturation Rate (%)
80
70
60
50
40
30
20
10
0
52.7%
25.5%
41.1%
52.7%
65.4%
Q1 (Low)
Q2
Q3
Q4 (High)

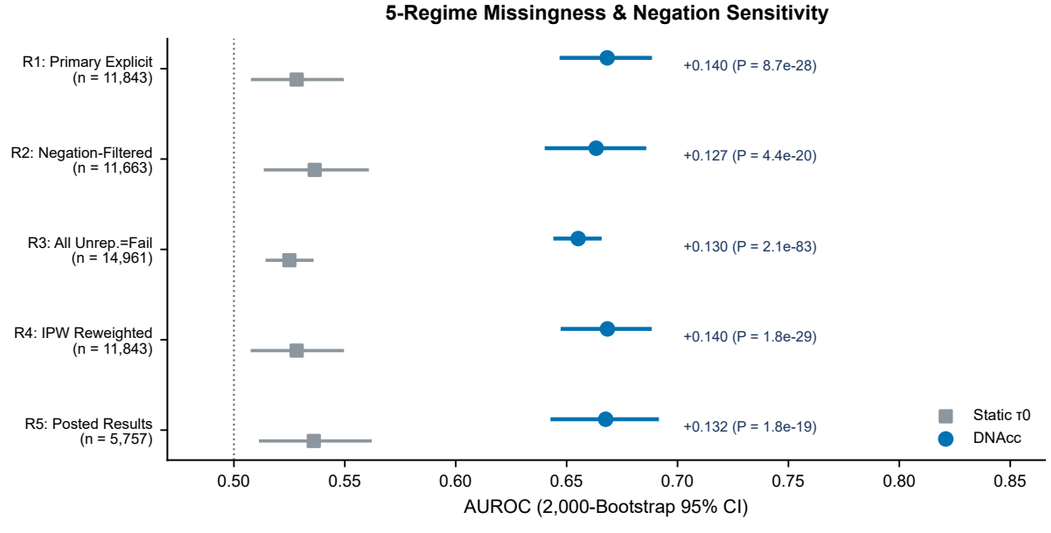
e


10-Decile Calibration (n = 23,967)
Ideal
Static τ0
DNAcc
Observed Rate
0.8
0.6
0.4
0.2
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Predicted Probability
Decision Curve Analysis (DCA)
Advance None
Advance All
Static τ0
DNAcc
Net Benefit
0.4
0.2
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Decision Threshold Probability

f


Clinical Target Remodeling Phase Space
Lineage Retained
Ectopic Lineage Switch
PCSK9
THRB
PDGFRB
PDCD1
TGFB1
LOXL2
Disease Remodeling Drift (Δτ = τ_late - τ_early)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.2
0.4
0.6
0.8
1.0
Healthy Baseline Tissue Specificity (τ0, HPA 154 Cell Types)

g


5-Regime Missingness & Negation Sensitivity
R1: Primary Explicit
(n = 11,843)
R2: Negation-Filtered
(n = 11,663)
R3: All Unrep.=Fail
(n = 14,961)
R4: IPW Reweighted
(n = 11,843)
R5: Posted Results
(n = 5,757)
+0.140 (P = 8.7e-28)
+0.127 (P = 4.4e-20)
+0.130 (P = 2.1e-83)
+0.140 (P = 1.8e-29)
+0.132 (P = 1.8e-19)
Static τ0
DNAcc
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
AUROC (2,000-Bootstrap 95% CI)